Sadeeq Adeyemi

Perception Engineer, Multi-Object Tracking and State Estimation

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SUMMARY

I build tracking systems that keep an object's identity while it is hidden from the camera, and I measure them on public benchmarks. Over the past year I have taken that one problem from simulation, to real driving data, to Meta's SAM 3 segmentation models. I work in Python, C++, PyTorch and ROS 2, and use AI coding agents daily.

SKILLS

Tracking and estimation: multi-object tracking and segmentation, Kalman filters (EKF, UKF), sensor fusion, data association (Hungarian, JPDA, MHT)
Perception: instance segmentation, SAM 3, YOLOv8, re-identification, lidar and camera calibration and projection, point clouds
Software: Python, C++17, PyTorch, ROS 2, OpenCV, NumPy, filterpy, MATLAB, Git, Docker, Linux, AWS
Evaluation: HOTA, IDF1 and MOTA with TrackEval; KITTI, KITTI-MOTS, MOTS20, MOT17, NVIDIA PhysicalAI-AV

EXPERIENCE

Robotics Research Assistant, University of Maine Robotics Laboratory 2025 – present
- Build mobile robots end to end, from sensor integration and wiring to ROS 2 nodes and lidar point-cloud perception.
- Implemented EKF localization fusing IMU, wheel odometry and lidar, with costmap planning and obstacle avoidance.

Materials Engineer, R.W. Gillespie, Maine 2024

Machine Learning Engineer and Structural Steel Fabricator, NewTech Steel, Nigeria 2018 – 2023
- Deployed a computer vision model for real-time fatigue and defect detection in steel; automating the test cut its time by 50%.
- Led fabrication and erection of structural steel for factory and auditorium projects.

RESEARCH

Tracking Through Occlusion, graduate research, University of Maine 2026
- Benchmarked SAM 3 and SAM 3.1 zero-shot with the official HOTA evaluation on an A100. On KITTI-MOTS val, cars scored 82.6 HOTA with SAM 3 and 74.6 with SAM 3.1. On MOTS20 train, SAM 3 kept a person's identity through 94% of 245 real occlusions, SAM 3.1 63%.
- Traced half of SAM 3.1's deficit to its shipped tracking settings by reading Meta's source code; a controlled run raised its car HOTA from 74.0 to 78.3.
- Added a Kalman-filter layer that gives a lost ID back when the object reappears: SAM 3.1's identity kept through occlusion rose from 63% to 70%.
- Mined 10,871 pedestrian occlusions from 1,256 NVIDIA driving clips with a ray-cast line-of-sight test, with lidar projected into the fisheye camera.
- On that real geometry, an occlusion-aware tracker kept identity through 1 to 5 second hides in 57% of 1,600 cases; a standard tracker kept 2%.
- Built a live 3D browser lab for occlusion experiments (46 automated tests), cross-checked against MathWorks GNN, JPDA and TOMHT trackers.

EDUCATION

MS, Data Science and Engineering, University of Maine 2025 – present

MS, Structural Engineering, University of Maine 2023 – 2025

MSc, Project Management, University of Portsmouth, United Kingdom 2023

BEng, Civil Engineering, Federal University of Technology Akure, Nigeria 2019

Certifications: AWS Certified Machine Learning Specialty; AWS Certified Solutions Architect, Associate